EV Network — Internal Operations Platform (Proposal 1)

Prepared by EvoFront · Oct 21, 2026 · @Mohan

Executive summary

EvoFront will build one internal operations platform (web + Android + iOS) that runs the client's 30+ EV charging stations from a single place, with every approval and payment release gated by the owner. It is delivered in 16 weeks and shares one backend with the Customer Charging & Payments App (Proposal 2).

Business context. The client operates 30+ charging stations with staff spread across sites. Running attendance, daily routines, maintenance, expenses and payouts across these sites calls for one system, full owner visibility and a single audit trail of who approved what.

What the platform delivers

- One live view of every station, every employee and every open task
- Structured daily routines (opening checks, charger health, cleanliness, cash/QR reconciliation) with photo and geo proof
- An owner-only approval desk: no expense, purchase, advance, salary or vendor payment is released without the owner's sign-off
- Role-based access (RBAC) so each person sees and does only what their role permits
- A full audit log of every request, approval, rejection and payout

Success measures

Measure	Target
Stations with daily checklist completed	95%+ of station-days
Approval turnaround (owner)	Under 24 hours median
Payments released without owner approval	Zero
Attendance captured digitally	100% of field staff

Users, roles and RBAC

Six default roles cover the network; the owner can create custom roles by combining permissions, and every permission is scoped to one station, a cluster of stations, or all stations.

Role	Who	Scope	Can do	Cannot do
Owner (Super Admin)	Client / promoter	All stations	Everything; sole approver for all money movement; manage roles	—
Operations Manager	Head office	All or assigned cluster	Assign staff and tasks, review reports, verify and recommend requests	Approve or release any payment
Accounts / Finance	Accountant	All stations	Prepare payouts, upload bills, reconcile, export to Tally	Approve; release funds without owner approval
Station Supervisor	One per station or cluster	Assigned stations	Run checklists, raise expense/purchase requests, mark staff attendance, log incidents	See other stations' data or finances
Field Technician	Maintenance staff	Assigned tickets	Accept and close maintenance tickets, log parts used, upload photos	Raise payments above a set limit
Station Attendant	Shift staff	Own station, own shift	Check in/out, complete shift checklist, raise issues, request advance or leave	View reports or other staff data

RBAC design principles

- Permission = action × module × scope (e.g. "approve · expenses · all stations")
- Money-moving permissions (approve, release) are locked to the Owner role and cannot be delegated without a second owner confirmation
- Optional delegate approver when the owner is unavailable, with a value cap and expiry date — owner's choice, off by default

- Every role change is recorded in the audit log

Feature modules

Twelve modules make up the platform; modules 1–8 are core (Phase 1) and 9–12 complete it (Phase 2).

#	Module	Key features	Phase
1	Station registry	Station profiles (address, geo-fence, chargers, connector types, capacity, landlord/lease details), cluster grouping, station status (live, under maintenance, closed)	1
2	Employee directory	Profiles, KYC documents, role, assigned station(s), shift pattern, salary structure, emergency contact, onboarding/offboarding checklist	1
3	Attendance and shifts	Geo-fenced check-in/out with selfie, shift roster, late/absent alerts, leave requests, overtime capture, monthly attendance sheet	1
4	Daily routines and checklists	Configurable checklists per station and shift (opening, charger health, cleanliness, safety, closing), photo and geo proof, missed-checklist alerts to supervisor and owner	1
5	Tasks and work orders	Assign one-off or recurring tasks, due dates, priority, comments, photo evidence, escalation when overdue	1
6	Maintenance and incidents	Charger fault tickets, preventive maintenance calendar, OEM/AMC vendor tracking, parts used, downtime log per charger, incident reports (safety, theft, damage)	1
7	Requests and approvals	Expense claims, petty cash, purchase requests, salary advances, vendor bills, leave — all routed to the owner (see next section)	1
8	Notifications	Push, SMS and WhatsApp alerts for approvals, missed checklists, faults, payouts	1

#	Module	Key features	Phase
9	Inventory and assets	Spares and consumables stock per station, asset register (chargers, transformers, CCTV), reorder alerts, asset transfer between stations	2
10	Payouts and payroll inputs	Payout queue for approved items, bank transfer file / payout API, salary sheet from attendance, advance recovery, payslips	2
11	Reports and owner dashboard	Station-wise opex, staff productivity, uptime, open tickets, approval ageing, revenue from Proposal 2 alongside costs; Excel/PDF export	2
12	Documents and compliance	Licences, electrical safety certificates, insurance, lease agreements with expiry reminders	2

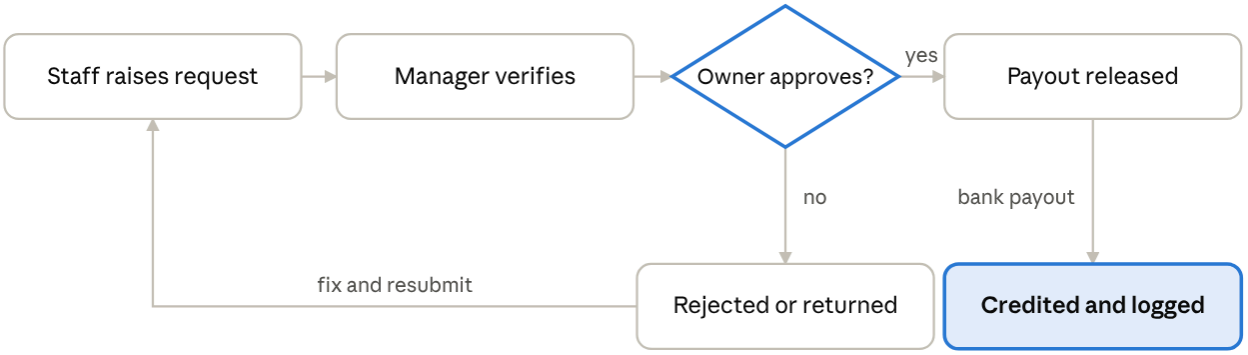
Mobile-first details for field staff

- Works on low-end Android phones; offline capture of checklists and photos, auto-sync when back online
- Telugu, Hindi and English interface
- One-tap actions on the home screen: Check in, Today's checklist, Raise issue, Request money

Owner-only approval and payment release

Every request that moves money ends at one desk: the owner's. Managers can verify and comment, Accounts can prepare, but only the owner approves, and only an approved item is released.

No payment is released without the owner approving it



approval and payment release flow · owner is the only approver

A rejected or returned request goes back to the person who raised it with the owner's remarks; an approved one is paid out and recorded against the station and employee.

Request types routed to the owner

- Expense claims and petty cash top-ups
- Purchase requests (spares, consumables, services)
- Vendor and AMC bills
- Salary advances and incentives
- Monthly salary release (approved as one batch)
- Refund overrides from the customer app (Proposal 2)

Approval desk features

- One inbox on web and mobile, sorted by value and age; bulk approve for routine items
- Each request shows the bill/photo, station budget used so far, requester history and the manager's remark
- Approve, reject or return with a comment; approval confirmed with PIN or biometric
- Reminders to the owner when items wait more than 24 hours
- Station-wise monthly budgets with alerts when 80% is used
- Released payments carry the bank reference (UTR) back into the record, so every rupee traces from request to credit

Platforms, architecture and security

The platform ships as a responsive web admin plus a single cross-platform mobile app for Android and iOS, sharing one backend with the customer charging app.

Platforms

Surface	Primary users	Technology
Web admin	Owner, Ops Manager, Accounts	React / Next.js
Mobile app (Android + iOS)	Owner (approvals on the go), Supervisors, Technicians, Attendants	Flutter — one codebase, published to Play Store and App Store
Backend	—	Modular Python FastAPI services, PostgreSQL, Redis, object storage for photos/documents

Integrations

- SMS (DLT-registered templates) and WhatsApp Business API for alerts
- Google Maps for geo-fencing and station maps
- Bank payout API (RazorpayX or the client's corporate banking API) or bank-upload file for releasing approved payments
- Tally export for accounting
- Read access to Proposal 2's transaction data for revenue-vs-cost reporting

Security and controls

- OTP + PIN login on mobile; device binding for staff phones
- Owner approvals need a second factor (PIN or biometric) at the moment of approval
- Immutable audit log: who did what, when, from which device and location
- Role and station-scoped data access enforced at the API, not just the screen
- Data hosted in India (AWS Mumbai region), encrypted at rest and in transit
- Personal data handled in line with the Digital Personal Data Protection Act, 2023

Non-functional targets

- 99.5% uptime for the backend
- Screens load in under 2 seconds on a 4G connection
- Sized for 30+ stations and 300+ staff at launch, scalable to 200+ stations

Scope and delivery

Phased delivery (16 weeks)

Phase	Weeks	Outcome
Discovery and UX	1–3	Signed-off requirements, RBAC matrix, approval rules, clickable prototype
Phase 1 build	4–10	Modules 1–8 on web and mobile; owner approval desk live
Phase 2 build	11–13	Modules 9–12, payouts, dashboards, Tally export
UAT and pilot	14–15	Pilot at 3 stations, fixes, staff training
Rollout	16	All 30+ stations live, Play Store and App Store release, handover

In scope

- Web admin, Android app, iOS app, backend and admin configuration for all 12 modules
- Data migration of stations and employees from the client's Excel sheets
- Training for owner, managers and supervisors; user guides in English
- 30 days of post-launch warranty support

Exclusions

- Full statutory payroll processing (PF, ESI, TDS filing) — the platform produces payroll inputs only
- Hardware: biometric devices, CCTV, phones
- Direct integration with charger hardware (covered in Proposal 2)
- Third-party charges (cloud hosting, SMS/WhatsApp messaging, payout API fees), billed to the client directly by the providers

Project terms

- The client provides one decision-maker for sign-offs within 3 working days
- Store developer accounts (Google Play, Apple) are opened in the client's name
- Approval rules and limits are finalised by the end of week 3

Inputs required from the client

- [ ] Headcount by role and station, current and planned for the next 12 months
- [ ] Revenue collection modes at each station (digital only, or cash as well)
- [ ] Bank account and payout method for releasing approved payments
- [ ] Delegate approval limits, if any low-value items are to be approved on the owner's behalf
- [ ] Accounting software in use (Tally or other)